EQUIPMENT NAME:	**FIM III Booster Station HV Switchboard**
EQUIPMENT NUMBER:	**F23-SW-601**

Procurement Package	"EL006" – HV Switchboard
Equipment No.	**Equipment Name**
F23-SW-601	FIM III Booster Station HV Switchboard
Description	Air Insulated (AIS) with a bus voltage transformer, bus earth switch, 400A withdrawable fused contactors, and 630A withdrawable circuit breakers. Bus earth switch shall be key interlocked with incomer circuit breaker to prevent closure with live bus.
Reference Documents	2000-F23-SLD-EL-40001 – FIM III Booster Station HV Switchboard Single Line Diagram 2000-F23-SLD-EL-40002 – FIM III Booster Station HV Switchboard Key Interlocking Single Line Diagram
Reference Specifications	2000-F00-STS-EL-10004 – HV Electrical Testing Specification 2000-F00-STS-EL-10015 – High Voltage Switchboards and Switchgear Specification 2000-F00-STS-GE-10001 – Packing and Shipping Specification 2000-F00-STS-GE-10002 – Site Specific Information 2000-F00-STS-GE-10003 – Plant Numbering Specification

Scope of Supply

Work Included

1. Design, supply, and fabrication of one (1) 11kV switchboard including, but not limited to:
   - Withdrawable type switchboard, single tiers
   - Air insulated switchboard
   - Rating: 1250A bus, 25kA 3 sec, AFLR
   - 48VDC auxiliary supply voltage
   - Earth switches are to be motorised and equipped with visual confirmation - by mirror and light for visual confirmation of earth switch positions
   - 1 x bus earth switch electrically interlocked to incomer circuit breakers
   - Shunt-trip, Close and No Volt Coils
   - Circuit Breaker Remote switching and racking capability (Remote switching panel to be supplied by Switchroom Vendor)
   - Fused Contactor Remote switching and racking capability (Remote switching panel to be supplied by Switchroom Vendor)
   - Earth switch (cable side) – motorised and to have remote operation capability
   - SE-330AU earthing resistor monitoring relay for NER
   - One set of Protection CT's for SEL 751#JP0G in each Incomer (including for arc flash).
   - Common CT specifications:
     - Protection current transformers - 5P20, 2.5 VA with burden to be appropriately sized by the Vendor
     - Metering current transformers - CL 0.2, 2.5 VA Metering current transformers with burden to be appropriately sized by the Vendor
   - Common VT specifications: 11/√3 / 0.11/√3 kV 15VA CL0.2 & 20VA CL 3P
   - Fortress keys – vendor to liaise with Fortress Distributor for key management.

2. All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment
   - Circuit breaker trolley for removing circuit breakers and fused contactors
   - Switching and/or racking handles
   - Internal arc absorbers - no external arc ducting required. Switchroom pressure calculation by others.
   - Equipment nameplates and labels
   - Bus bar connection pieces
   - Inter-panel wiring
   - Any other items which are normally suppled with this equipment

3. Supply of all necessary documentation, type test certificates (no type test repetition), drawings and test documentation. Vendor to refer to Vendor Drawing and Document Requirements List below.
4. Routine testing at manufacturer's premises.
5. Packaging suitable for sea and road transport to Perth-based switchroom vendor premises.
6. Interlocking locks and keys (Fortress), including:
   - Incomers - single set (incl. barrel) Fortress and keyed alike to upstream 33kV switchgear. Fortress locks to be managed by Fortress distributor. Earth switch should be interlocked to upstream circuit breaker.
   - VSD feeders – single key on fused contactor earth switch to key exchange box (key exchange box is for VSD power section doors, to ensure cable earth switch is on prior to opening VSD power section doors).

7. **F75-SW-601 FIM III Booster Station HV Switchboard (drg. 2000-F23-SLD-EL-40001)**

A. Equipment List:

Panel No.	Description	Rating (Amps)	Protection Relay	Protection CT Ratio	Metering CT Ratio	Core Balance CT Ratio
1	Bus VT and Earth Switch	NA	NA	NA	NA	NA
2	F23-PP-651A-VSD FIM III Booster Pump 1 HV VSD – *Fused Contactor*	400		100/1	NA	50/1

3	F23-PP-651B-VSD FIM III Booster Pump 2 HV VSD – *Fused Contactor*	400		100/1	NA	50/1
4	Incomer – Circuit Breaker - *Additional core of 300/1A CTs for SEL735#L54R*	630	Littelfuse SE-330AU (NER Monitor)	300/1	300/1	NA
5	F23-TR-602 FIM III Booster Station 11kV/0.433kV 2.5MVA Transformer – *Circuit Breaker*	630		100/1	NA	50/1
6	SPARE – *Circuit Breaker*	630		100/1	NA	50/1

Work Excluded

- Cabling external to switchboard;
- Spare parts
- Offloading the equipment at vendor premises
- Installation and commissioning at site.

	Unit	Duty	Vendor	Notes	Rev
General					
Switchgear Type		Air Insulated (AIS)	Air Insulated (AIS)	3	0
Access required		Single sided, front. Manufacturer to state if rear access is required for cable terminations or maintenance e.g. VT access	Single sided, front	1	0
Internal Arc Fault Containment		AFLR	AFLR	1	0
Partition Class		PM	PM	1	0
Service Continuity Category		LSC2A	LSC2B	1	0
Intended Installation		Elevated Switchroom, Front Access. Vendor to state if rear access is required	Front access	3	0
Arc vents/chutes		Top Chimneys	Top chimneys (No external room venting required)	3	0
Make		ABB	ABB	3	0
Model		ZS1	ZS1	3	0
Country of Manufacture			China	2	0
Compliance with Australian Standards (or equivalent IEC standards)	Comply/Do Not Comply	Comply to IEC62271	Yes / IEC62271	3	0
Drive and Load Descriptions		As per Single Line Diagram	Noted	1	0
CB ratings		As per Single Line Diagram	Noted	1,5	0
Environmental Conditions					
Plant Location		Western Australia	Noted	1	0
Service Location	Indoor/Outdoor	Indoor (State any applicable deratings)	Indoor	1	0
Site Data		Refer to 2000-F00-STS-GE-10002 – Site Specific Information	Noted	1	0
Supply					
Number of phases		3	3	1	0
Input Voltage	kV	11 ± 10%	11	1	0
Frequency	Hz	50 ± 2%	50	1	0
Neutral Earthing		Impedance (limited to 25A) by Neutral Earthing Resistor	Noted	1	0
Switchboard Ratings					
Rated Voltage	kV	12	12	1	0
Rated Busbar Current	A	1250	1250	1	0
Rated Power Frequency Withstand Voltage	kV rms	28	28	1	0
Rated Lighting Impulse Voltage	kV peak	75	75	1	0
Rated Short time Withstand Current and Time	kA/s	25kA/3s	25kA/3 sec	3	0
Rated Internal Arc Fault Containment	kA/s	21kA/1sec	25kA/1 sec	3	0
Rated Peak Withstand Current	kA	62.5	62.5kA	3	0
Endurance Class	Comply/Do Not Comply	E2, M2, C2	Comply	1	0
Busbar Arrangement	Single/Duplicate	Single	Single	1	0
Busbar Identification	Yes/No	Yes	Yes	1	0
Insulated Busbars	Yes/No	Yes	Yes	1	0

	Unit	Duty	Vendor	Notes	Rev
Busbar Material		Copper	Copper	3	0
Future Busbar Extension at Either End	Yes/No	Yes (Both)	Yes (both)	3	0
Earth Bar Withstand Current and Time	kA/s	16kA/1s	Comply	3	0
Control Voltage	V	48VDC	48V DC	3	0
Mechanical Interlocks					
Bus Earth Switch / Incoming Circuit Breakers	Comply/Do Not Comply	All incomer circuit breakers Fortress key interlocked with bus earth switch. Refer to single line diagrams	Comply	1	0
Incoming Earth Switch / Upstream Circuit Breaker	Comply/Do Not Comply	Incoming earth switches key interlocked with upstream disconnector. Refer to single line diagrams	Comply	1	0
Cable Cubicle Access	Comply/Do Not Comply	Cubicle cannot be opened when earth switch is opened	Comply	1	0
Circuit Breaker / Earth Switch	Comply/Do Not Comply	Earth switch and FC/CB cannot be In Service and Closed at the same time. Refer to single line diagrams	Comply	1	0
Visual position confirmation	Comply/Do Not Comply	CB truck position and earth switch to be visible to confirm open and earth switch ON/OFF operation (window or camera)	For earthing switch On/Off visual confirmation, Mirror+light is considered, however it is dependent on the power cable terminations, which may hinder the viewing.	1	0
Fused Contactor Earth switch	Comply/Do Not Comply	Fortress key interlock required between earth switch and VSD compartment.	Comply	1	0
Incoming Circuit Breakers (Incomers)					
Manufacturer		ABB	ABB	3	0
Model		VD4/P	VD4/P	3	0
Circuit Breaker Type		VCB, Withdrawable	VCB, Withdrawable	3	0
Rated Voltage	kV	12	12kV	1	0
Rated Nominal Current	A	630	630	3	0
Rated Short time Withstand Current and Time	kA/s	25kA/3s	25kA /3s	3	0
Making Current – Peak	kA	63	63	3	0
Breaking Current – Symmetrical / Asymmetrical	kA	25	25	3	0
Close Coil Voltage	V	48VDC	48VDC	1	0
Close Coil Power Demand (energise, latched)	W	200W	200W, 5W	3	0
Tripping Method		Shunt Trip and Undervoltage Release	Shunt Trip and Undervoltage Release	3	0
Shunt Trip Coil Voltage	V	48VDC	48VDC	1	0
Shunt Trip Coil Power Demand (energise, latched)	W		200W, 5W	2	0
Undervoltage Coil Voltage	V	48VDC	48VDC	1	0
Undervoltage Coil Power Demand (energise, latched)	W		200W, 5W	2	0
Spring Charging Motor Supply Voltage	V	48VDC	48VDC	1	0
Spring Charge Power Demand (energise, latched)	W		600W, 200W	2	0
Number of Permissible Circuit Breaker Operations					
At Rated Current	No.	E2	E2	3	0
At Rated Fault Current	No.	E2	E2	3	0
CB Close Time	ms	50…80	50…80	3	0
CB Open Time	ms	33…60	33…60	3	0
Number of Auxiliary Contacts		5NO+5NC	5NO+5NC	3	0
Trip Circuit Supervision Relay	Yes/No	As per SEL's relay part references/functions offered	As per SEL's relay part references/functions offered (excluded from ABB Scope)	3	0
Earth Switch Remote Operation	Yes/No	Yes	Yes	3	0
Fused Contactors					
Manufacturer			ABB	2	0
Model			VSC	2	0
Fused Contactor Type		Withdrawable	Withdrawable	3	0
Rated Voltage	kV	12	12	1	0
Rated Nominal Current with fuse	A	160	160	3	0

	Unit	Duty	Vendor	Notes	Rev
Rated Short time Withstand Current and Time	kA/s	25kA/3s	25kA/3s	3	0
Making Current – Peak	kA		50kA Value linked to the breaking capacity of the fuse. Refer to the fuse manufacturer's documentation	2	0
Breaking Current – Symmetrical / Asymmetrical	kA		50kA Depending on the capacity of the coordinated fuse	2	0
Close Coil Voltage	V	48VDC	48VDC	1	0
Fuse contactor burden	W		TBC	2	0
Number of operations			2,000,000	2	0
Current Transformers					
Phase Current Transformers					
Manufacturer & Model		ABB	ABB	3	0
Type	Block/ Toroid	Block	Block	3	0
Duty		Overcurrent Protection & Metering (Incomers) Overcurrent and SEF Protection (Feeders)	Comply	1,6	0
Ratios		Refer to SLDs All outgoing feeders shall have 50/1 core balanced CTs	As per offer technical specs., 50/1 core balance CTs are included	3,6	0
Class		Protection 5P20 2.5VA Metering CL0.2 2.5VA	Comply	3,6	0
Voltage Transformer					
Manufacturer & Model		ABB	ABB	3	0
Type			TBA	2	0
Ratios		11kV/√3 / 110V/√3	11kV/√3 / 110V/√3	3	0
Rating		15VA & 20VA	Comply	3	0
Class		CL0.2 & CL 3P	Comply	3	0
Fixed or Withdrawable		Fixed for Riser panel, Withdrawable for Bus-VT panels	Withdrawable for Bus-VT panel	3	0
Cubicle Details					
Ingress Protection Rating		IP4x	IP4x	3	0
Cubicle Control Voltage	V	48VDC	48V DC	1	0
Local Indication LED Lamps					
Supply Voltage	V	48VDC	48V DC	1	0
Type		Semaphore Type	Comply	1	0
Required Indication		Circuit Breaker Status Earth Switch Status Interlock Status	Comply, with hard-wired electrical interlocks	1	0
3 Phase Cable Side Capacitive Phase Indicator	Yes/No	Yes	Yes	1	0
Anticondensation Heater Supply	V	240VAC	240V AC	3	0
Anticondensation Heater Power	W	100W	100W	3	0
Electric Controls					
Local/Remote Selector	Yes/No	Yes	Yes	1	0
Local Trip	Yes/No	Yes	Yes	1	0
Local Close	Yes/No	Yes	Yes	1	0
Remote Control Interface (Hardwire to Airlock)					
Remote Control Selected	Yes/No	Yes, Dry Contact	Comply	1	0
Earth Switch Open/Closed	Yes/No	Yes, Dry Contact (Changeover)	Comply	1	0
Circuit Breaker Open/Closed	Yes/No	Yes, Dry Contact (Changeover)	Comply	1	0
Interlock Status	Yes/No	Yes, Dry Contact	Comply as per detailed engineering design	1	0
Circuit Breaker Open	Yes/No	Yes, Remote 48V Signal		1	0
Circuit Breaker Close	Yes/No	Yes, Remote 48V Signal		1	0
Circuit Breaker/Fused Contractor truck position	Yes/No	Yes, Racked In and Racked Out		1	0
External Paint Colour		RAL7035	RAL7035	3	0
Internal Paint Colour		Unpainted High Quality Glossy Finish (ALUZINC)	Unpainted High Quality Glossy Finish (ALUZINC)	3	0
Paint Specification			As attached	2	0
Viewing Windows/Ports/Cameras					
Type of device to enable visual confirmation of earth switch	Window/ Camera	Window with mirror and light. No camera	Window with mirror & light (No camera)	3	0

	Unit	Duty	Vendor	Notes	Rev
Window/Camera to view Bus Earth Switch On/Off	Yes/No	Window with mirror and light. No camera	Window with mirror & light (No camera)	3	0
Window to view Cubicle Earth Switch On/Off	Yes/No	Window with mirror and light. No camera	Window with mirror & light (No camera)	3	0
Light to Assist Viewing Earth Switch	Yes/No	Window with mirror and light. No camera	Window with mirror & light (No camera)	3	0
Maintenance Access					
Access to VT	Front/Rear/ Top	Front	Front	3	0
Access to Busbar Joints	Front/Rear/ Top	Top	Top	3	0
Access to Maintenance Connections	Front/Rear/ Top	Front	Front	3	0
Access to Cable Connections	Front/Rear/ Top	Front	Front	3	0
Cable Connections					
Cable Entry	Top/Bottom	Bottom	Bottom	3	0
Cable Connection Type	Type C/ Flags	Refer to ZS1 IOM	Refer to ZS1 IOM	3	0
Incoming Cable Type		XLPE	Noted	1	0
Maximum Cable Size	mm^2	2 x 3C 120mm^2	Comply	1	0
Outgoing Cable Type		XLPE	Noted	1	0
Maximum Cable Size	mm^2	1 x 3C 150mm^2	Comply	1	0
Recommended Termination Kit			TBA	2	0
Gland Plate Window Size (L x W)	mm x mm		Pls. refer attached drawings	2	0
Height from Gland Plate to Cable Terminals	mm	Approximately 500	~500mm approx.	3	0
Phase and Wire Colours					
Phase Colours	Comply/Do Not Comply	Red, White, Blue	Comply	1	0
Neutral Colour	Comply/Do Not Comply	Black	Comply	1	0
Earth Colour	Comply/Do Not Comply	Green/Yellow	Comply	1	0
48V DC Tripping and closing (+ve)	Comply/Do Not Comply	Grey	Comply	1	0
48V DC Tripping and closing (-ve)	Comply/Do Not Comply	Black with white tracing	Black	1	0
Labelling Required					
Switchboard Label		Etched Traffolyte label, centrally located, with switchboard equipment number, description and SLD number at minimum	ABB standard Gravoply labels are considered.	1	0
Switchboard/Tier Nameplate Label		Etched Traffolyte switchboard nameplate label centrally located. Each tier shall have a nameplate label relevant to the tier contents		1	0
Tier/Cubicle Label		Etched Traffolyte label per tier, with tier/cubicle number, service description and fed from location (on incoming cubicle) at minimum		1	0
Interlock Details		Etched Traffolyte label per interlock, detailing interlock type and procedure to place interlock in service / out of service		1	0
Other Cubicle Labelling		Etched Traffolyte label per item (eg. Protection relay, indicator lamp, switches) detailing applicable equipment number and description		1	0
Earth Switch and Circuit Breaker Isolation Points		Etched Traffolyte label at each isolation point		1	0
Test Requirements					
Type		AS62271-200 Sub Clause 6.1 (or equivalent)	Type test certified	3,6	0
Routine		AS62271-200 Clause 7 (or equivalent)	Manufacturer standard routine tests based on IEC 62271-200	3	0
Special		N//A		1	0
Shipping and Delivery					
Packaging		Suitable for sea and road freight	Suitable for sea freight	3	0
Shipping Section Dimensions			TBA	2	0
Weight if Individual Panel/Tier			As mentioned in technical offer	2	0

	Unit	Duty	Vendor	Notes	Rev
Overall Dimensions of Switchboard Assembly			As mentioned in technical offer	2	0
Number of Shipping Sections			TBA	2	0

Notes
1. Fixed. The Vendor shall comply with this data, as a minimum. Vendor shall state their applicable ratings/data or compliance
2. Vendor to Advise
3. Preferred, Vendor to advise and provide reasoning for difference
4. Type test certificates to be provided with tender documentation
5. Tender documentation shall include detailed single line diagram which is reflective of the Vendor's offer
6. Vendor shall advise whether multiple CTs or single CT with two secondary windings shall be utilised for incomer modules.

Vendor Drawing and Document Requirements List (VDDR)

E – Electronic Copy H – Hard Copy MDR – Manufacture's Data Report

FAT – Factory Acceptance Testing IOM – Installation and Operating Manual

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
A		**Production Schedule**				
	A01	Production Schedule / Bar Chart		1E	2	
	A02	Monthly Report		Monthly		
B		**Registers and General Drawing Requirements**				
	B01	Drawing Schedule / Register (Drawings Expected)		1E	4	
	B02	General Arrangement with Foundation Loads (in AutoCAD 2010 and PDF)		1E	6-8	1E
	B03	Cross-section		1E	6-8	1E
C		**Mounting and Erection Drawings**				
	C01	Foundation and Mounting Details (Provided in B02)		1E	6-8	1E
	C02	Pipe Spools				
	C03	Outline Dimensions, Cut-out and Support Details (Provided in B02)		1E	6-8	1E
	C04	Match Marked Erection Drawings				
	C05	Motor GA (with Fixing and Shaft Details, and Overall Dimensions)				
D		**Electrical Drawings (in electronic form as appropriate)**				
	D01	Single Line Diagrams		1E	6-8	1E
	D02	Schematic Drawings		1E	10-12	1E
	D03	Terminations / Loop Diagrams		1E	10-12	1E
	D04	Panel Layout and Fabrication Details		1E	6-8	1E
	D05	Nameplate		1E	6-8	1E
	D06	Cable Block Drawings				
	D07	Electrical and Instrumentation GAs		1E	6-8	1E
E		**Process and Piping Drawings**				
	E01	Process Flow Sheets				
	E02	Process and Instrument Diagram				
	E03	Pneumatic or Hydraulic Diagram				
	E04	Lubrication Diagram				
F		**Technical Data**				
	F01	Equipment Information Schedule	1E			
	F02	Electrical Performance data sheet		1E	6-8	
	F03	Equipment List (Part of Schematics)		1E	6-8	1E
	F04	Vendor's Surface Protection Details		1E	6-8	
	F05	Lubrication Data Schedule				
	F06	Material Schedule				
	F07	Bolt Schedule				
	F08	Fan Performance Curves				
	F09	Pump Performance Curves (with Duty Points Marked)				
	F10	Cable Schedule				
	F11	Vendor Catalogue Information	1E			
	F12	Calculations				
	F13	Completed $_{\text{Engineer}}$ Datasheets	1E			
	F14	Motor Datasheets				
	F15	Electrical Equipment List				
	F16	Instrument List				
	F17	I/O List (Part of Schematics)		1E	10-12	

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
	F18	Communications Addressing and Signal Mapping List				
G		**Inspection, Certification and MDR**				
	G01	Inspection and Test Plans		1E	6	
	G02	Inspection and Test Procedures (generic)				
	G03	Quality Plans and Procedures (generic)		1E	4-6	
	G04	Manufacturer's Data Report (MDR) – Index / Contents		1E	4-6	
	G05	Manufacturer's Data Report (MDR)		1E	Delivery +2	
	G06	Material Certificates				
	G07	Statutory Authority Certificates				
	G08	Test Data and Other Certificates		In MDR		
	G09	Welding Procedure / Specification				
	G10	Non-Destructive Testing (NDT) Procedures				
	G11	Material Safety Data Sheets				
	G12	Commissioning Plan and Procedure				
	G13	Structural Design Certification				
	G14	Performance Guarantee Statement				
H		**Shipping and Packing Details**				
	H01	Material / Shipping List	1E	1E	Delivery -4	
	H02	Package Type and Quantity (Carton, Case, Crate)		1E	Delivery -2	
	H03	Shipping Packages – Weights and Dimensions		1E	Delivery -2	
	H04	Shipping and Handling Procedure		1E	12	
J		**Manuals**				
	J01	Installation, Operation and Maintenance Manual (Preliminary)		1E	Delivery -4	
	J02	Installation, Operation and Maintenance Manual (Final)		1E	Delivery	
K		**Spare Parts Information**				
	K01	Spare Parts Catalogue and Recommended Spares List (Copy also in Manuals)	1E	1E	8-10	
	K02	Recommended Spare Parts List with Pricing (Not Included in Manuals)	1E	1E	8-10	
	K03	Special Tools list	1E	1E	8	
M		**General**				
	M01	Control Philosophy and Logic Diagram				
	M02	Functional Description				